

ReSolve: Verified Mechanical CAD Edits and Linked Vector Drawings

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Abstract

001 Editing a mechanical drawing is useful only when the
002 underlying part changes as requested and the delivered
003 views remain consistent. We study a CAD-first work-
004 flow that predicts a validated program patch, executes
005 the edited CadQuery model, and exports linked SVG
006 views. Our main experiment uses published Bench-
007 CAD edit pairs with a local family- and template-
008 component-disjoint train/validation/test split. A fresh
009 1.5B QLoRA model is trained on 492 retained ed-
010 its and evaluated on all 126 retained held-out cases,
011 using source-program equality and the author’s re-
012 leased STEP solids. The adapter yields 77/126 ap-
013 plicable patches and 12 exact target programs; 12
014 of 125 valid released STEP references are matched
015 strictly. A subsequent three-epoch H100 LoRA run
016 on the same retained split produces 61/126 applicable
017 patches and 15/125 strict solid matches. We report ex-
018 ecutable solids and volume agreement separately, with
019 a format-relaxed base-model control. A smaller, ex-
020 plicitly localized pilot provides 150 verified edit pairs
021 and paired drawing sheets. Separate deterministic
022 truss and plate controls demonstrate axial FEM equi-
023 librium and clearance violations; those results are not
024 attributed to the learned CAD model. The study mea-
025 sures code-conditioned editing and linked view gener-
026 ation, not drawing perception or general engineering
027 safety. BenchCAD examples lack a physical analysis
028 contract, so no FEM validation is claimed for their me-
029 chanical edits.

030 1. Introduction

031 A request to remove a bracket flange, enlarge a bore or
032 add a shaft slot must change more than a caption. The
033 edited part should preserve unrelated features, remain
034 executable, and produce coherent views. If the request
035 concerns structural performance, it additionally needs

material, units, supports, loads and a declared phys- 036
ical model. Shape validity alone cannot supply those 037
assumptions. 038

This paper focuses on mechanical CAD edits and 039
their linked vector drawings. Figure 1 illustrates two 040
published reference pairs: removing a flange while re- 041
taining the twisted web and first-flange holes, and 042
adding an obround blind slot to a stepped shaft. These 043
require compound three-dimensional geometry rather 044
than a symbol substitution. The figure shows released 045
references, not model predictions. 046

We implement a CAD-first edit-execute-check 047
route. A language model reads an executable pro- 048
gram and the user’s instruction and returns a source- 049
line patch. A validating executor applies the patch to 050
the original program. CadQuery builds the resulting 051
solid; orthographic and isometric SVG views are ex- 052
ported from that solid. Independent scoring compares 053
the executed result with the author’s STEP reference. 054
A malformed patch, invalid solid and valid but incor- 055
rect edit are distinct outcomes. 056

Our main data comes from an online release rather 057
than only locally manufactured requests: BenchCAD’s 058
published edit pairs [13]. Because its edit subset is 059
originally a held-out benchmark, using it for training 060
changes the protocol. We explicitly construct a local 061
split before training, hold out complete family/tem- 062
plate components and start from fresh base weights. 063
We do not report an official BenchCAD leaderboard 064
result. 065

Contributions and scope. 066

- A reproducible local training/evaluation protocol 067
over published mechanical edit pairs, with pinned 068
provenance, family/template isolation, disclosed to- 069
ken exclusions and every retained test case scored. 070
- A validated CAD-program patch interface and de- 071
terministic linked SVG export, evaluated with both 072
program and executed-solid agreement. The learned 073
model receives code; it is not a drawing-perception 074

model.

- A separate 150-pair hard-part pilot with before/after STEP solids and four-view drawing sheets, plus a transport control showing why patch-format gains must not be equated with engineering reasoning.
- Explicit separation of mechanical geometry checks, deterministic structural FEM equilibrium and plate clearance rules. No mechanical safety or broad method-novelty claim follows from these small, single-seed experiments.

2. Related work

Executable CAD and editing. DeepCAD [11] and Text2CAD [4] model parametric construction histories. CAD-Editor [12] studies instruction-based sequence editing. BenchCAD [13] supplies executable mechanical programs and published edit pairs, while neuralCAD-Edit [7] evaluates expert-requested edits. Our experiment reuses released supervision; a program-patch interface and execution check are implementation choices, not evidence that CAD editing itself is new.

Drawings and vector graphics. Drawing2CAD [8] links vector views to CAD histories. DeepSVG [1] and StarVector [9] model vector graphics, and SVGE-ditBench [6] benchmarks instruction-based SVG edits. Our learned experiment conditions on CAD code and exports SVG after execution. It therefore does not establish SVG-to-CAD extraction, image understanding, or robust annotation recovery from manufacturing sheets.

Physical evaluation. FEM-Bench [5] tests scientific code generation. CADEngBench [10] evaluates functional CAD edits and matched linear-static FEA. Geometry and FEA are complementary: the latter requires an analysis contract. We retain an explicitly specified axial-truss control but do not fabricate loads or materials for BenchCAD parts. A public runnable CADEngBench release was not verified in this work, so its reported FEA results are related work rather than our reproduced benchmark.

3. Mechanical edit–execute–check workflow

3.1. Program and patch contract

Let C be a CadQuery program and u an edit instruction. The model predicts a patch p , and the executor produces $\hat{C} = \mathcal{E}(C, p)$. Each patch operation contains a zero-based original line index, a deletion count and

a list of replacement lines. All coordinates refer to the original source, not an intermediate revision. Operations must be ordered, non-overlapping, within bounds and well typed. Parsing the executed program provides a second syntax check.

Training targets are line diffs between the published source and edited programs after removing comments and empty lines. Comment normalization avoids supervising changes to the upstream parameter-comment block as geometric edits. A round-trip check requires that applying every target patch reproduces its normalized target exactly. The online instructions are preserved; source-line answers are not added to the prompt.

3.2. Execution and vector views

CadQuery executes the patched program to a boundary-representation solid $\hat{B}$. The scorer requires valid geometry, at least one solid and positive volume. Downloaded programs run through a restricted executor with permitted imports, no private-attribute access or file-I/O builtins, and a 45-second per-case subprocess budget. This restriction is defense in depth, not a general Python sandbox. Unsupported constructs, invalid programs and timeouts are recorded separately.

For the verified drawing pilot, front, top, right and isometric SVG projections of both original and edited solids are saved alongside STEP exports. A before/after sheet binds these views to the request and geometry measurements. The projections share a three-dimensional source; they are not independently generated views. They are CAD-derived line drawings with edit annotations, not fully dimensioned production blueprints. Automatic projection scaling can differ between views, so rendered pixel size is not a physical measurement.

3.3. Independent solid scoring

The online test reference is the author’s released STEP solid B^* , pinned by content hash. We report volumetric intersection-over-union,

$$\text{IoU}(\hat{B}, B^*) = \frac{V(\hat{B} \cap B^*)}{V(\hat{B}) + V(B^*) - V(\hat{B} \cap B^*)}. \quad (1)$$

Our local thresholds are 0.95 for broad volume agreement and 0.99999 for strict agreement. Neither is presented as an official BenchCAD acceptance rule. Exact normalized-program AST equality is complementary: IoU may overlook a small but important feature, and multiple programs can produce the same shape. The current scorer does not certify feature-specific tolerances, preserved constraints, assembly joints, dimen-

Published BenchCAD source / reference edit pairs

twisted bracket | Remove the second flange; retain the twisted web and first-flange holes.

SOURCE

RELEASED TARGET



stepped shaft | Add a centered blind obround slot to the upper shaft face.

SOURCE

RELEASED TARGET



Reference illustrations only. No stress or FEM result is inferred from these shapes. BenchCAD, CC-BY-4.0.

Figure 1. Published mechanical CAD edit pairs held out in the local protocol. Top: remove the perpendicular end flange while preserving the twisted web and the first-flange holes. Bottom: add a centered blind obround slot to the upper cylindrical shaft section. These are source/released-target illustrations, not learned model predictions. Both views are exported from solids; shape appearance supplies no stress or FEM result.

sions, standards compliance or manufacturing suitability.

Verification is post-edit evaluation. The QLoRA loss is supervised token prediction, not FEM or solver feedback; the experiment does not evaluate iterative repair after a failed check.

4. Online dataset and split protocol

Published pairs. We use the `edit-bench` configuration of BenchCAD/BenchCAD, pinned at revision 5919f578ab09ec283603a082fab07c7639ab56eb. It contains 748 instruction/source/target pairs with original and target STEP data [13]. The dataset declares CC-BY-4.0. Our downloaded parquet checksum and source-record identifiers are retained. Pipe, duct, fitting, manifold, valve, hose, resistor, circuit and PCB family names are excluded, removing 38 records and leaving 710 mechanical edit pairs.

Isolation before training. A complete mechanical family is one initial group. Families whose source or target ASTs share a template after masking numeric literals are joined into a component. Hash-ranked components are assigned approximately 70/10/20 to training, val-

idation and test before normalization, patches or augmentation. This gives 533/50/127 pairs. Family and component identities are disjoint across splits; no target from a held-out family enters fine-tuning. Template isolation is conservative but does not prove all cross-dataset geometric ancestry is disjoint. The fresh run does not load the earlier mechanical adapter.

Context-limited coverage. We retain a pair only when its full chat, including target, fits 3,072 tokens and its target patch fits 512 tokens. The rule is applied once before baseline evaluation and training. It retains 492/44/126 train/validation/test pairs, excluding 48 records across all splits. No example is silently truncated. The one excluded test pair and all excluded training examples remain identified in the run manifest. Results apply to the retained slice rather than the entire release.

Protocol change. The author labels this edit subset a held-out benchmark. We deliberately re-partition it into local training and evaluation data to satisfy the present study’s training objective. This invalidates claims of official benchmark-test generalization or direct leaderboard comparability. The local held-

215 out components remain unused for optimization and
216 checkpoint selection.

217 Reference-quality limits. Readback finds one invalid
218 rivet STEP among the 126 retained test references.
219 We also identify a concrete instruction/target con-
220 flict: `topup_stepped_shaft_top_slot_rev` requests
221 cutting a blind slot, but its source already has the slot
222 and its target removes the operation. Published la-
223 bels are retained without silently repairing them; this
224 case and its source diff are recorded. The experiment
225 measures agreement with released targets, not certi-
226 fied compliance with every natural-language request.
227 We do not infer that every reverse-edit record has this
228 defect.

229 Earlier drawing pilot. A separate sample of 2,000
230 BenchCAD code-generation parts yielded 150 verified
231 local numeric-edit pairs from 76 hard parts and 40 fam-
232 ilies, partitioned 108/22/20 by family. It requires at
233 least eight CAD calls and compound geometry, and
234 excludes external edit-program templates after mask-
235 ing literals. Chamfer, hole, extrusion and fillet edits
236 contribute 62, 48, 24 and 16 pairs. The sample exports
237 904 referenced SVG views and 150 before/after sheets.
238 Its instructions disclose the edit line and numeric an-
239 swer; it is a transport feasibility pilot, not the main
240 online benchmark.

241 5. Online editing experiments

242 Training. We start from a fresh 4-bit Qwen2.5-
243 Coder-1.5B-Instruct conversion [3] and train a QLoRA
244 adapter [2] using MLX on an Apple M5 with
245 16 GB memory. The model conversion revision is
246 `b3252a2f97102b1fb1571fec2c9b27219a8536be`. Seed
247 17, batch size one, learning rate 10^{-4} and the last eight
248 transformer blocks are fixed. Adapters have rank 8,
249 scale 20, no dropout, and cover attention and MLP
250 linear projections in those blocks. Prompt tokens are
251 masked from the loss. We run 492 optimizer steps, cor-
252 responding to one pass of the retained training rows,
253 and use the final checkpoint. Validation is logged, not
254 used to select among test-informed alternatives. Saved
255 adapters are reloaded before evaluation.

256 Evaluation and metrics. Base and trained models re-
257 ceive the same original programs, published instruc-
258 tions and patch contract. Greedy decoding is capped
259 at 512 new tokens. All 126 retained test records are
260 evaluated in the same fixed order. Applicability re-
261 quires a valid patch and syntactically valid program;
262 AST equality requires the executed program to match

the published target’s normalized syntax. Solid scoring
executes the actual prediction and compares it with the
released STEP reference. No gold edits repair a model
prediction.

Transport control. The base model frequently emits
a bare list of edits rather than the required object con-
taining `edits`. We therefore add a deterministic diag-
nostic that wraps a syntactically valid bare list while
preserving every operation. It makes 56 of 126 base
predictions applicable, but none matches the target
AST. This control distinguishes one schema mismatch
from the remaining content and coordinate errors; it
does not rewrite geometry, infer missing operations or
correct indexing.

Results and operational coverage. The trained model
produces 77/126 applicable patches and 12/126 exact
target ASTs. Of 125 independently valid STEP refer-
ences, 39 predictions build valid solids, 18 reach $\text{IoU} \geq 0.95$,
and 12 reach strict $\text{IoU} \geq 0.99999$ (9.6%). The bare-list
base control builds 3 solids, with 1 broad and 0 strict
matches. Thus the experiment exposes a substantial
remaining gap: most held-out edits fail to produce the
reference geometry, particularly beyond literal-parameter
changes. Independent reference readback finds 125/126
valid target solids. One released rivet reference is in-
valid and is excluded from the geometric success denomi-
nator, while all task outputs remain reported. For the
trained run, 39 cases are geometrically scored, 37 have
execution errors and 49 have invalid patches; there are
0 timeouts and 0 Boolean-comparison errors. The ob-
served code failures include invalid feature operations
and damaged source coordinates; they are distinct from
valid solids with low IoU . We keep base and trained
outputs, patch errors, executed-solid measurements,
reference hashes and category breakdowns. One seed
and one model size cannot establish a robust scaling
law or broad superiority over CAD agents. No official
benchmark claim or statistical population failure rate
is inferred from this local slice.

Additional H100 run. After the local experiment,
available server compute supports a separately re-
ported CUDA run. A fresh unquantized BF16 Qwen2.5-
Coder-1.5B-Instruct model is trained on the identical
492 retained training rows for three epochs (1,476
updates) on one NVIDIA H100 NVL. PEFT adapters
use rank 8, alpha 160 (scale 20), no dropout and the
same seven attention/MLP projections in the last eight
blocks. Seed, batch size, learning rate, prompt mask-
ing, context limit and greedy generation limit remain
unchanged. We reload the final adapter

System	Applicable	Target AST	IoU ≥ 0.95	IoU ≥ 0.99999
Base, strict interface	0	0	0	0
Base, bare-list transport control	56	0	1	0
Fresh online-pair QLoRA	77	12	18	12

Table 1. Local online-pair benchmark: counts out of 126. All systems use the same family/template holdout. The transport control only wraps a bare JSON edit list into the required object, without changing coordinates or content. Timeouts and comparison errors remain separately disclosed; they are not measured incorrect geometry. This is a local split, not an official BenchCAD score.

Category	Cases	Target AST	Strict match
T1	31	11	11
T2	18	1	1
T3	18	0	0
T4	34	0	0
T5	25	0	0

Table 2. Trained results by published edit category. Category IDs are retained from the source; counts are not pooled across different evaluation contracts.

and evaluate all 126 cases; no test-based checkpoint selection is performed. Different precision, software implementations and epoch counts prevent interpreting the comparison as an isolated hardware effect or an independent preregistered confirmation.

The final CUDA adapter yields 61/126 applicable patches and 16/126 target AST matches. Against 125 valid released STEP solids, 36 predictions execute, 19 meet broad IoU and 15 meet strict IoU (12.0%). There are 65 invalid patches, 24 execution errors, 0 timeouts and 0 comparison errors; the same invalid released reference is excluded from the geometry denominator.

Localized drawing pilot. The earlier 100-step adapter retained 92 training, 22 validation and 12 test examples after context filtering. All 12 held-out exact patches execute and match their references. Under a unique-substring line parser the base model already matches 11/12 solids. That result mainly improves transport formatting and cannot support a claim of advanced CAD reasoning. The online experiment uses published instructions without line hints, a fresh adapter and its own held-out families.

6. Separate structural and clearance controls

The CAD benchmark above checks mechanical shape, not FEM. We retain a separately identified deterministic SVG route to show what physical or manufacturing checks require when a drawing does supply a contract. These synthetic controls are not training data or accuracy evidence for the learned BenchCAD model.

Axial trusses. An importer reads supported untagged truss geometry, section dimensions, material notes, nodal loads and supports. A bounded request interpreter generates validated SVG tree patches. Re-importing the actual patched SVG determines member lengths, orientations and axial areas. Linear-elastic pin-jointed elements use EA/L stiffness; assembled displacements, stresses and reactions are recomputed. The numerical verdict checks equilibrium under small-displacement and nodal-load assumptions, not buckling, fatigue, joints or material strength.

Plate clearances. A separate importer recovers a rectangular boundary and circular-hole centres, diameters, thickness and stated limits. It computes overlap, edge distance and ligament. A faithful hole-enlargement request can reduce clearance below the printed minimum. A geometric violation is an engineering consequence of a correctly executed request, not necessarily an edit error; no continuum plate stress follows from a clearance score.

For example, increasing all truss member depths from 20 to 40 mm changes the recovered area and reduces peak axial stress from 11.861 to 5.930 MPa under the same load. Enlarging plate holes from 10 to 16 mm at fixed centres 20 mm from a boundary reduces edge clearance from 15 to 12 mm and fails a 15 mm rule. These are distinct checks with explicitly stated assumptions. The saved audit and replayed source/edited SVGs remain separate from the new online training run.

When FEM is applicable. A stress, stiffness, buckling, thermal or contact request needs physical units, constitutive properties, loads and boundary conditions, followed by solver and mesh checks. The inspected BenchCAD edit pairs do not provide this analysis contract. We mark their FEM status not evaluated, rather than inventing loads and presenting an arbitrary synthetic simulation as verification of the released part.

System	Applicable	Target AST	IoU ≥ 0.95	IoU ≥ 0.99999
CUDA base	0	0	0	0
CUDA base, list wrapper	47	0	0	0
CUDA final LoRA	61	16	19	15

Table 3. Additional three-epoch H100 experiment. Counts cover the same 126 retained tasks and 125 valid references, using the unchanged local geometry thresholds. Both base and final model are decoded with the CUDA implementation; the wrapper control preserves predicted edit contents.

Route	Edits	Outcome
Axial trusses	20,008	Equilibrium passes
Perforated plates	19,992	699 clearance violations

Table 4. Historical structural pipeline audit. All 40,000 supported edits match their stored target trees. These counts describe a deterministic synthetic pipeline, not a learned CAD benchmark. The plate violations are faithful edits flagged by the geometric checker.

Verification coverage. All six illustrated H100 cases are actual learned CAD program edits, followed by execution and comparison with released STEP targets. None has FEM verification. The online benchmark and localized mechanical pilot check geometry; the plate controls check clearances; only the separate axial-truss controls solve a finite-element equilibrium problem. Geometry agreement does not establish stress, stiffness or safety. FEM for a mechanical part would require an explicitly specified material, unit system, load case, supports and a validated mesh/solver setup, which are absent from these released edit pairs.

7. Limitations and conclusion

This is a small, single-seed code-conditioned CAD editing study. The online supervision comes from a synthetic mechanical benchmark re-partitioned locally; family/template checks do not prove complete geometric-ancestry isolation. A documented instruction/target conflict and one invalid STEP reference show that published labels require further auditing. Context filtering removes some of the longest parts and edit patches. Predicted code is limited by the executor and its time budget, and the current comparison checks solid volume agreement rather than feature-specific engineering tolerances. Our SVG views are deterministic projections; no learned drawing perception, annotation binding or production-sheet reconstruction is evaluated.

The structural controls are deterministic and use bounded grammars. Their equilibrium and clearance results cannot be transferred to arbitrary CAD shapes

or interpreted as general safety certification. The mechanical benchmark has no FEM claim because its necessary physical specifications are absent.

The completed workflow links published CAD edit supervision, validated program patches, executed solids and vector views. By reporting schema adherence, target syntax, actual solid agreement and operational failures separately, it makes the supported outcome reviewable. Stronger evidence requires larger contexts, multiple seeds and models, feature-preservation metrics, genuine drawing-conditioned editing and physical tasks with independently specified analysis contracts.

References

- [1] Alexandre Carlier, Martin Danelljan, Alexandre Alahi, and Radu Timofte. DeepSVG: A hierarchical generative network for vector graphics animation. In Advances in Neural Information Processing Systems, 2020. 2
- [2] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In International Conference on Learning Representations, 2022. 4
- [3] Binyuan Hui et al. Qwen2.5-coder technical report, 2024. 4
- [4] Mohammad Sadil Khan et al. Text2CAD: Generating sequential CAD designs from beginner-to-expert level text prompts. In Advances in Neural Information Processing Systems, 2024. 2
- [5] Saeed Mohammadzadeh, Erfan Hamdi, Joel Shor, and Emma Lejeune. Fem-bench: A structured scientific reasoning benchmark for evaluating code-generating llms, 2025. 2
- [6] Kunato Nishina and Yusuke Matsui. Svgeditbench v2: A benchmark for instruction-based svg editing, 2025. 2
- [7] Toby Perrett, Matthew Bouchard, and William McCarthy. neuralcad-edit: An expert benchmark for multimodal-instructed 3d cad model editing, 2026. 2
- [8] Feiwei Qin, Shichao Lu, Junhao Hou, Changmiao Wang, Meie Fang, and Ligang Liu. Drawing2cad: Sequence-to-sequence learning for cad generation from vector drawings, 2025. 2

- [9] Juan A. Rodriguez et al. StarVector: Generating scalable vector graphics code from images and text. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025. 2
- [10] Harmanjot Singh, Abhra Dubey, and Jorge Alejandro Amador Herrera. Cadengbench: It looks like cad, but does it work? evaluating parametric design, assembly reasoning, and physics simulation, 2026. 2
- [11] Rundi Wu, Chang Xiao, and Changxi Zheng. DeepCAD: A deep generative network for computer-aided design models. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2021. 2
- [12] Yu Yuan, Shizhao Sun, Qi Liu, and Jiang Bian. Cad-editor: A locate-then-infill framework with automated training data synthesis for text-based cad editing, 2025. 2
- [13] Haozhe Zhang, Kaichen Liu, Miaomiao Chen, Lei Li, Shaojie Yang, Cheng Peng, and Hanjie Chen. Benchcad: A comprehensive, industry-standard benchmark for programmatic cad, 2026. 1, 2, 3

477 A. Reproduction and evidence contracts

478 Online source. The `edit-bench` parquet is down-
479 loaded from [https://huggingface.co/datasets/](https://huggingface.co/datasets/BenchCAD/BenchCAD)
480 [BenchCAD/BenchCAD](https://huggingface.co/datasets/BenchCAD/BenchCAD) at the revision stated in Sec. 4.
481 The original published split name is `edit_bench`. Its
482 748 rows are not silently treated as 748 independent
483 source shapes; grouping uses all original and edited
484 programs before split assignment. Numeric literals are
485 masked for template linkage; all records of a linked
486 component retain one split.

```
487 {"edits": [  
488   {"start": 5, "delete": 1,  
489    "insert": ["    .hole(6.0)"]}]  
490 ]}
```

491 The zero-based line coordinate addresses the original
492 code. A replacement list is required even for one line.
493 Insertion uses zero deletions. The executor rejects over-
494 lapping operations and embedded newline characters.
495 This example describes the interface, not an online gold
496 answer.
497

498 Saved artifacts. The online dataset manifest pins the
499 parquet checksum, split checksums, source IDs, cat-
500 egories and reference STEP hashes. The run mani-
501 fest pins model revision, package versions, seed, opti-
502 mization command, retained rows and omissions. The
503 baseline and reloaded final adapter use identical test
504 prompts. Predictions remain unmodified; the bare-list
505 transport control is saved as a separate diagnostic. Ge-
506 ometry scores include every retained case, including
507 invalid patches, code errors, unsupported constructs,
508 timeouts and Boolean-comparison errors.

509 Evidence separation. The 150-pair localized pilot,
510 710-pair online protocol and historical 40,000-edit
511 structural audit are different datasets and contracts.
512 Their results are never pooled. The localized pilot’s
513 family test contains only 12 context-retained examples
514 and gives line/number hints. The online experiment
515 is trained afresh with original natural-language re-
516 quests and evaluates 126 context-retained cases. Struc-
517 tural FEM is only established for specified pin-jointed
518 trusses.

519 Validation. Regression checks test multi-operation
520 original-coordinate execution, insertion/deletion, mal-
521 formed indexing, overlap rejection and comment
522 preservation. Dataset checks require train/valida-
523 tion/test family and template-component disjointness,
524 exact target-patch replay and pinned split hashes. Fi-
525 nal adapters are reloaded for generation and checked

for finite tensors. Reference STEP readback and all
output geometry scores are retained. Source/refer-
ence illustrations are labeled as such and do not depict
model predictions.

B. Held-out model edit examples

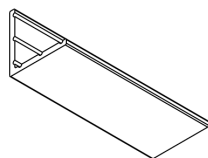
Figures 2 and 3 show six actual predictions from the fi-
nal three-epoch H100 adapter, alongside the input solid
and released STEP target. These are illustrative cases
selected after evaluation, not a random sample or addi-
tional test set; all 126 evaluated cases remain included
in the reported aggregate results. We select three strict
matches and three executable but geometrically incor-
rect edits to expose both supported behavior and re-
maining limitations. Each view is independently fitted
to its panel, so apparent drawing size is not a dimen-
sional comparison. Instructions and measured volume
IoU accompany every row. The shapes provide no ma-
terial, load, stress or FEM evidence.

Held-out H100 CAD edits: strict matches

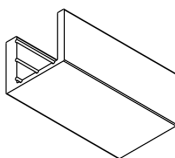
mounting angle box x f130 | T1 | IoU 1.00000

Increase the box X dimension from 33.0 to 42.9.

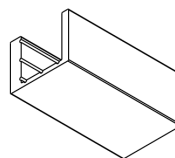
SOURCE



RELEASED TARGET



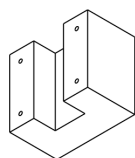
H100 PREDICTION



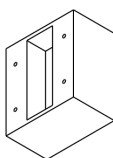
locator block box y f130 | T1 | IoU 1.00000

Increase the box Y dimension from 82.2 to 106.86.

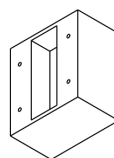
SOURCE



RELEASED TARGET



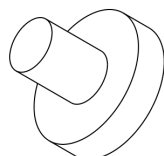
H100 PREDICTION



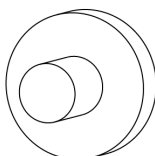
t3medplus stepped shaft rot diag60 | T2 | IoU 1.00000

Rotate the entire part by 60 degrees about the axis (1, 0, 1).

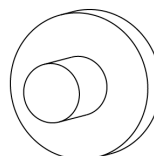
SOURCE



RELEASED TARGET



H100 PREDICTION



Views fit independently; use measured IoU, not apparent drawing size. BenchCAD CC-BY-4.0. No FEM claim.

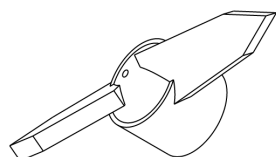
Figure 2. Three successful held-out H100 edits. Rows show a mounting-angle dimension change, locator-block width change and a 60-degree stepped-shaft rotation. Columns contain the source, author’s released target and actual final-adaptor prediction. Each meets strict volume IoU ≥ 0.99999 . These examples demonstrate reference agreement within the disclosed code-conditioned protocol; they do not certify general engineering correctness.

Held-out H100 CAD edits: partial / failed edits

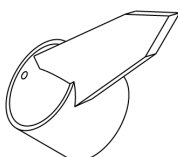
propeller bore widen compute | T4 | IoU 0.63155

Remove one of the two propeller blades; keep the hub, spinner, and the remaining blade.

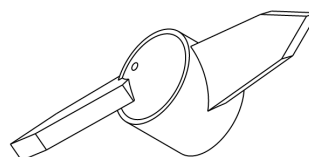
SOURCE



RELEASED TARGET



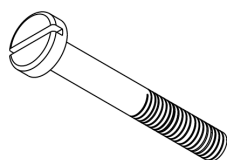
H100 PREDICTION



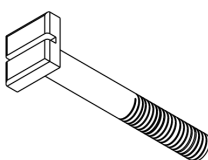
t5hard pan head screw U a | T5 | IoU 0.58449

Replace the round pan head of the screw with a square head of the same across-flats width and the same thickness, keeping the threaded shaft, the screwdriver slot, and the rounded top edge unchanged.

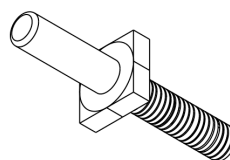
SOURCE



RELEASED TARGET



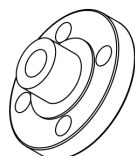
H100 PREDICTION



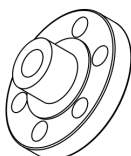
t5hard round flange V a | T5 | IoU 0.92018

Change the ring of 4 polar mounting through-passages on the flange face to a ring of 6 mounting through-passages of the same diameter, evenly spaced around the central axis at the same radius.

SOURCE



RELEASED TARGET



H100 PREDICTION



Views fit independently; use measured IoU, not apparent drawing size. BenchCAD CC-BY-4.0. No FEM claim.

Figure 3. Three executable but incorrect held-out H100 edits. Removing a propeller blade, replacing a screw head and changing a flange mounting-hole pattern yield valid solids but fail the strict reference threshold. Their volume IoUs are approximately 0.632, 0.584 and 0.920. Source, released target and model output remain separate; no reference operations repair these predictions. Small feature errors can remain important even when bulk volume overlap is high.